0.1 Copmactness

Definition 0.1.1. A Topological Space X is said to be *compact* if: Every open cover contains a finite subcover. i.e.,

$$\text{If } X = \bigcup_{\alpha \in \Lambda} \mathcal{U}_\alpha,\ (\mathcal{U}_\alpha \in \mathcal{T}), \text{ then there is finite subcover such that } X = \bigcup_{i=1}^{N} \mathcal{U}_{\alpha_i}$$

This is equivalent with:

$$\text{If } \emptyset = \bigcap_{\alpha \in \Lambda} \mathcal{C}_\alpha,\ (\mathcal{C}_\alpha \text{ closed}), \text{ then there is finite subset such that } \emptyset = \bigcap_{i=1}^{N} \mathcal{C}_{\alpha_i}$$

Definition 0.1.2. Let X be a set. We say that a collection $\mathcal{A} \subset \mathcal{P}(X)$ satisfies *finite intersection property* if:

Every finite intersection of elements in $\mathcal{A}$ is Not empty.

Example. There are collections which satisfies FIP, but infinite intersection is empty.

1. $X = \mathbb{R}$, and let $A = \{(n, \infty) \mid n \in \mathbb{N}\}$.
2. $X = \mathbb{R}$, and let $A = \{\mathbb{R} \setminus F \mid F \text{ is finite set}\}$.

Theorem 0.1.3. Let X be a Topological Space. TFAE:

a) X is Compact Space.

b) If A is a collection of closed subsets of X satisfying *FIP*, then $\bigcap_{\mathcal{C} \in A} \mathcal{C} \neq \emptyset$.

c) If A is a collection of subsets of X satisfying *FIP*, then $\bigcap_{S \in A} \overline{S} \neq \emptyset$.

Proof. $a) \implies b)$. Proof by Contradiction:
Suppose that $A \subset \mathcal{P}(X)$ is a collection of closed subsets such that *FIP*. Assume that $\bigcap_{\mathcal{C} \in A} \mathcal{C} = \emptyset$.

$$\emptyset = \bigcap_{\mathcal{C} \in A} \mathcal{C} \textit{ if and only if } X = \bigcup_{\mathcal{C} \in A} (X \setminus \mathcal{C})$$

Since $X \setminus \mathcal{C}$ is open, there is a finite subcover by Compactness:

$$X = \bigcup_{i=1}^{N} (X \setminus \mathcal{C}_i) \textit{ if and only if } \emptyset = \bigcap_{i=1}^{N} \mathcal{C}$$

This is Contradiction with A satisfies *FIP*.
$b) \implies a)$. Proof by Contraposition:
Suppose that X is not Compact. Then, there exists an Open Cover $\mathcal{O}$ with no finite subcover: i.e.,

$$X = \bigcup_{\mathcal{U} \in \mathcal{O}} \mathcal{U} \textit{ if and only if } \emptyset = \bigcap_{\mathcal{U} \in \mathcal{O}} (X \setminus \mathcal{U})$$

And,

$$\text{For any finite subset of } \mathcal{O},\ F = \{\mathcal{U}_i \mid i = 1, \ldots, N\} \text{ satisfies } X \supsetneq \bigcup_{i=1}^{N} \mathcal{U}_i \textit{ if and only if } \emptyset \neq \bigcap_{i=1}^{N} (X \setminus \mathcal{U}_i)$$

Thus, $\mathcal{K} = \{X \setminus \mathcal{U} \mid \mathcal{U} \in \mathcal{O}\}$ satisfies *FIP*, but $\emptyset = \bigcap_{\mathcal{U} \in \mathcal{O}} (X \setminus \mathcal{U}) = \bigcap_{\mathcal{C} \in \mathcal{K}} \mathcal{C}$. Thus, not $a)$ implies not $b)$.
$c) \implies b)$ is trivial. $b) \implies c)$ is also trivial, because

$$\bigcap_{S \in A} S \subseteq \bigcap_{S \in A} \overline{S}$$

□

Theorem 0.1.4. Let X be a Compact Space, Y be a Topological Space.
If $f : X \to Y$ is Continuous Map, then $f[X]$ is Compact.

Proof. Let $\mathcal{O}$ be an open cover of $f[X]$. i.e, $f[X] \subset \bigcup_{\mathcal{U} \in \mathcal{O}} \mathcal{U}$. Now,

$$X \subset f^{-1}[f[X]] \subset f^{-1}\left[\bigcup_{\mathcal{U} \in \mathcal{O}} \mathcal{U}\right] = \bigcup_{\mathcal{U} \in \mathcal{O}} \underbrace{f^{-1}[\mathcal{U}]}_{\text{open, } f \text{ conti.}}$$

Since X is compact, there is a finite subcover such that

$$X \subset \bigcup_{i=1}^{N} f^{-1}[\mathcal{U}_i]$$

Consequently,

$$f[X] \subset f\left[\bigcup_{i=1}^{N} f^{-1}[\mathcal{U}_i]\right] = \bigcup_{i=1}^{N} f\left[f^{-1}[\mathcal{U}_i]\right] \subset \bigcup_{i=1}^{N} \mathcal{U}_i$$

□

Theorem 0.1.5. Closed set of compact space is compact.

Proof. Let X be a compact, and $E \subset X$ be a closed subset. Let $\mathcal{O}$ be an open over of E. Then,

$$X = E \cup (X \setminus E) \subset \left(\bigcup_{\mathcal{U} \in \mathcal{O}} \mathcal{U}\right) \cup (X \setminus E)$$

is an open cover of X. Thus, there is a finite subcover such that

$$X = \left(\bigcup_{i=1}^{N} \mathcal{U}_i\right) \cup (X \setminus E) \iff E \subset \bigcup_{i=1}^{N} \mathcal{U}_i$$

□

Theorem 0.1.6. Let X be a Topological Space, and β be a basis of X. Then, TFAE:

a) X is Compact Space.

b) Every open cover consisting of basis elements has a finite subcover.

Proof. $a) \implies b)$. Clear by definition of Compact.
$b) \implies a)$. Let $\{\mathcal{U}_\alpha \mid \alpha \in \Lambda\}$ be an Open cover of X. That is,

$$X = \bigcup_{\alpha \in \Lambda} \mathcal{U}_\alpha = \bigcup_{\alpha \in \Lambda} \bigcup_{\gamma \in \Gamma_\alpha} B_\alpha^\gamma$$

where $\{B_\alpha^\gamma \mid \gamma \in \Gamma_\alpha\}$ is subset of basis such that $\bigcup_{\gamma \in \Gamma_\alpha} B_\alpha^\gamma = \mathcal{U}_\alpha$. Now, by $2)$, there is finite subcover such that

$$X = \bigcup_{i=1}^{n} \bigcup_{j=1}^{m} B_{\alpha_i}^{\gamma_j} \subset \bigcup_{i=1}^{n} \mathcal{U}_{\alpha_i}$$

Thus, $\{\mathcal{U}_{\alpha_i} \mid i = 1, 2, \ldots, n\}$ is a finite subcover, we desired. □

Theorem 0.1.7. Let X, Y be Topological Spaces. Then, TFAE:

a) $X \times Y$ is Compact.

b) X and Y both are Compact.

Proof. $a) \implies b)$ is clear since projection preserves Compactness.
$b) \implies a)$ Let $\mathcal{O} \overset{\text{def}}{=} \{U \times V \mid U \in \mathcal{T}_X, V \in \mathcal{T}_Y\}$ be an Open cover of $X \times Y$.
Let $x \in X$ fix. Then, $\{x\} \times Y$ be a Compact, being $\{x\} \times Y \cong Y$ by Homeomorphism given by Projection.
Then, there is a finite subcover of $\mathcal{O}$ such that

$$\{x\} \times Y \subset \bigcup_{i=1}^{n_x} (U_i^x \times V_i^x)$$

Now, for each $x \in X$, define $U^x \overset{\text{def}}{=} \bigcap_{i=1}^{n_x} U_i^x$. Then, U^x is an open set containing x, and for any $i = 1, 2, \ldots, n_x$, $U^x \subset U_i^x$.
Since $\{U^x \mid x \in X\}$ be an open cover of X, there is a finite subcover such that

$$X = \bigcup_{i=1}^{m} U^{x_i}$$

being X is Compact. Now,

$$X \times Y = \left(\bigcup_{i=1}^{m} U^{x_i} \right) \times Y = \bigcup_{i=1}^{m} (U^{x_i} \times Y) \subset \bigcup_{i=1}^{m} \bigcup_{j=1}^{n_{x_i}} (U_j^{x_i} \times V_j^{x_i})$$

Thus, $\{U_j^{x_i} \times V_j^{x_i} \mid i = 1, 2, \ldots, m, \ j = 1, 2, \ldots, n_{x_i}\}$ be a finite subcover. □

Tube Lemma

Let X be a Topological Space, and Y be a Compact Space.
Then, for the product space $X \times Y$, and fixed $x_0 \in X$, the following statement holds:

For any open $N \subset X \times Y$ with $\{x_0\} \times Y \subset N$, there is an open $W \in \mathcal{T}_X$ such that $\{x_0\} \times Y \subset W \times Y \subset N$.

Proof. Clearly, $\{x_0\} \times Y$ compact, being $\{x_0\} \times Y \simeq Y$.
For any $y \in Y$, $(x_0, y) \in \{x_0\} \times Y \subset N$, thus there exist opens $U_y \in \mathcal{T}_X$ and $V_y \in \mathcal{T}_Y$ such that $(x_0, y) \in U_y \times V_y \subset N$.
Now, Clearly $\{U_y \times V_y \subset X \times Y \mid y \in Y\}$ be an open cover of $\{x_0\} \times Y$, thus there is a finite subcover such that

$$\{x_0\} \times Y \subset \bigcup_{i=1}^{N} (U_{y_i} \times V_{y_i}) \subset N$$

Set $W = \bigcap_{i=1}^{N} U_{y_i}$. Then, clearly $x_0 \in W$, and

Let $(x, y) \in W \times Y$. Then, since $Y = \bigcup_{i=1}^{n} V_{y_i}$, there is $1 \leq k \leq n$ such that $y \in V_{y_k}$.
Thus, $(x, y) \in U_{y_k} \times V_{y_k} \subset N$, this implies $W \times Y \subset N$. □

Theorem 0.1.8. Let Y be a Compact Space. Then, the following statements are true, but their converses are false:

1. If X be a Lindelöf Space, then the product Topology $X \times Y$ be a Lindel$\ddot{o}$f Space.
2. If X be a Countable Compact Space, then the product Topology $X \times Y$ be a Countable Compact Space.

Proof. 1. Let $\mathcal{O}$ be an open cover of $X \times Y$.
For any $x \in X$, $\{x\} \times Y$ is compact set, being $\{x\} \times Y \simeq Y$. Thus, there is a finite subcover of $\mathcal{O}$ such that

$$\{x\} \times Y \subset \bigcup_{j=1}^{N_x} U_j^x \quad (U_j^x \in \mathcal{O})$$

Since Tube Lemma, there is an open $W_x \in \mathcal{T}_X$ such that

$$\{x\} \times Y \subset W_x \times Y \subset \bigcup_{j=1}^{N_x} U_j^x$$

Meanwhile, since X is Lindel$\ddot{o}$f, therefore for an open cover $\{W_x \mid x \in X\}$
there exists a Countable subcover such that

$$X \subset \bigcup_{i=1}^{\infty} W_{x_i}$$

Consequently,

$$X \times Y \subset \left(\bigcup_{i=1}^{\infty} W_{x_i}\right) \times Y \subset \bigcup_{i=1}^{\infty} (W_{x_i} \times Y) \subset \bigcup_{i=1}^{\infty} \bigcup_{j=1}^{N_{x_i}} U_j^{x_i}$$

Now, $\left\{U_j^{x_i} \mid i \in \mathbb{N}, 1 \leq j \leq N_{x_i}\right\} \subset \mathcal{O}$ be a Countable Open Cover of $X \times Y$. □

Proof. 2. Let $\{U_n \subset X \times Y \mid n \in \mathbb{N}\}$ be a Countable open cover of $X \times Y$. For each finite subset $F \subset \mathbb{N}$, define

$$V_F \overset{\text{def}}{=} \left\{ x \in X \;\middle|\; \{x\} \times Y \subset \bigcup_{n \in F} U_n \right\}$$

Then V_F satisfies:
1) V_F is open: Let a finite subset $F \subset \mathbb{N}$ fix. For each $x \in V_F$, $\{x\} \times Y \subset \bigcup_{n \in F} U_n$ by definition.

Then, there is an open $W_x \in \mathcal{T}_X$ such that $\{x\} \times Y \subset W_x \times Y \subset \bigcup_{n \in F} U_n$ by Tube Lemma.

Meanwhile, $W_x \subset V_F$ because for all $s \in W_x$, $\{s\} \times Y \subset W_x \times Y \subset \bigcup_{n \in F} U_n$, thus $s \in V_F$.
In summary, for any $x \in V_F$, there is an open $W_x \in \mathcal{T}_X$ such that $x \in W_x \subset V_F$. Consequently, V_F is open of X.
2) $\{V_F \mid F \subset \mathbb{N}, |F| < \infty\}$ is a Countable Open Cover of X:
Countability given by above set is collection of subsets of Countable set. Meanwhile,
For any $x \in X$, there is a finite subcover of $\{U_n \mid n \in \mathbb{N}\}$ such that $\{x\} \times Y \subset \bigcup_{n \in F} U_n$ where F finite.

That is, $x \in V_F$. Now, the open cover of X,

$$\{V_{F_x} \mid x \in X\} \subset \{V_F \mid F \subset \mathbb{N}\}$$

at most Countable. Since X is Countably Compact Space, there is a finite subcover such that

$$X \subset \bigcup_{i=1}^{N} V_{F_i}$$

Consequently,

$$X \times Y \subset \left(\bigcup_{i=1}^{N} V_{F_i}\right) \times Y = \bigcup_{i=1}^{N} (V_{F_i} \times Y) \subset \bigcup_{i=1}^{N} \bigcup_{n \in F_i} U_n$$

That is, $\{U_i \mid i = 1, 2, \ldots, N,\ n \in F_i\}$ be a finite subcover. □